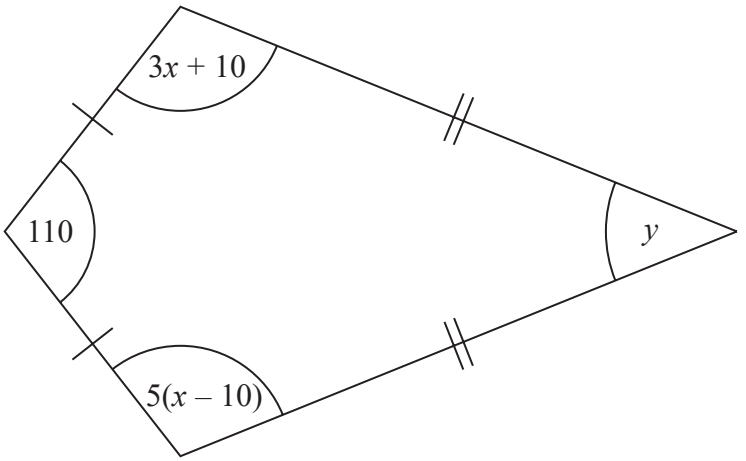
23 Here is a kite.



All angles are measured in degrees.

Work out the value of y.

1 mark
Equate equal angles

$3x + 10 = 5(x - 10)$: opposite angles of a kite.

×	x	-10
5	$5x$	-50

$5(x - 10) = 5x - 50$.

1 mark
Expand and isolate x

$3x + 10 = 5x - 50$ (−3x, −3x)

$10 = 2x - 50$ (+50, +50)

$60 = 2x$ (÷2, ÷2)

$x = 30$.

1 mark
Substitute x=30

Both equal angles $= 3 \times 30 + 10 = 100°$.

1 mark
50°

$y = 360° - 110° - 100° - 100° = 50°$.

(Total for Question 23 is 4 marks)